Midterm Report

ASTRO 320 - Midterm

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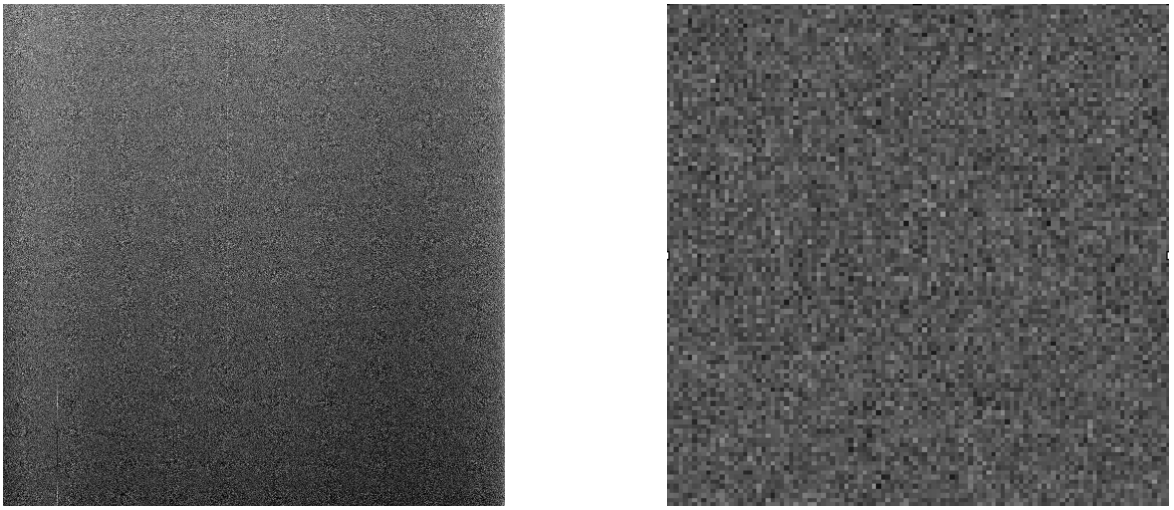
1 Exercise A

1.1 Introduction

Regardless of whether or not a Charge Coupled Device (CCD) is exposed to light, two sources of noise will be added to an image taken. One of them being the read noise and the other being dark current. The point of this exercise was to measure these two sources of noise. We began with a closed, dark dome, using single exposure mode and Astro 320 Dark as our exposure preset we took 7 dark exposures with varying length (0.1s-60s) all with X-Binning of 1. We then moved on to a constant exposure time with varying temperatures, taking 7 dark exposures of five minutes with different temperatures. Lastly, using the coldest temperature we were able to achieve (-34.6°C) we took our 20 0.12s bias frames.

1.2 Using Python Code to Find Read Noise

Making use of the two Python scripts provided, the first script measures the readout noise of the entire frame to be 9.637 DN (Figure 1a). Using the second script, we measured the readout noise of a 110×110 pixel defect-free region centered at (x = 2,300, y = 2,030) to be 9.636 DN (Figure 1b).



(a) Full bias frame.

(b) The 110×110 pixel defect-free region.

Figure 1: Regions used to measure the readout noise with the two Python scripts.

The two measurements differ by only 0.001 DN, which is not a statistically significant difference. To see whether that is even meaningful, we can estimate how precise the ROI measurement is. Each pixel's standard deviation is calculated from only 20 frames, so to get its scatter from pixel to pixel, we use

$$\sigma_s \approx \frac{\sigma}{\sqrt{2(N-1)}} \approx \frac{9.64}{\sqrt{38}} \approx 1.6 \text{ DN}$$

Averaging over the 110^2 = 12,100 pixels in the ROI brings the uncertainty on the mean down to about $1.6/\sqrt{12{,}100} \approx 0.014$ DN. The 0.001 DN difference is more than ten times smaller than this, so the two values agree completely within the uncertainty.

The two values likely agree because both scripts compare each pixel only with itself across the 20 bias frames, so fixed defects like the gradient, dark columns, and the bright top left edge in the master bias cancel out either way. This means choosing a defect free region doesn't remove anything the full frame measurement was picking up, though the specific region we chose may also play a role. The agreement also shows the read noise is uniform across the chip.

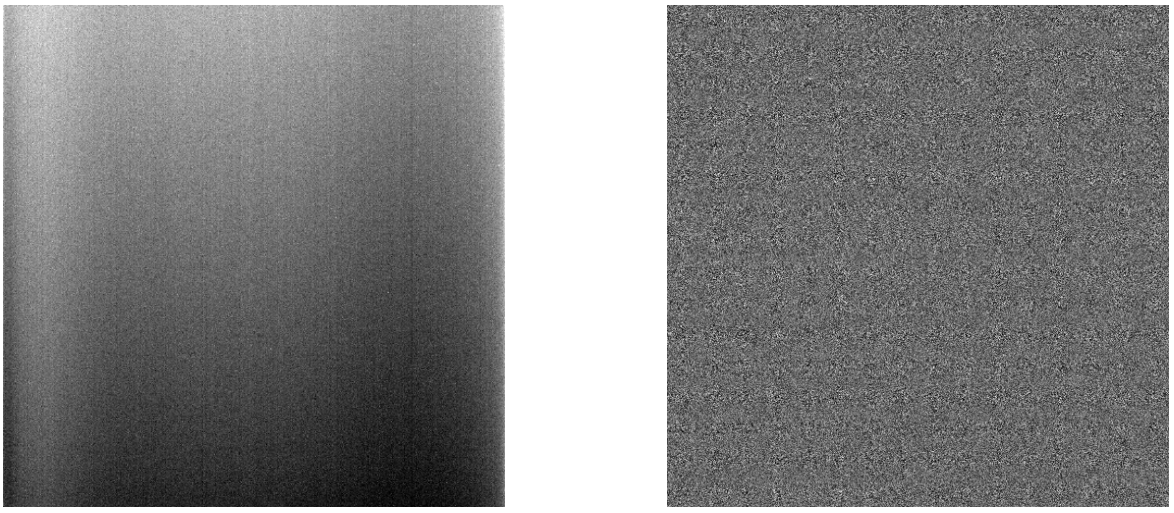
Image	Bias Frame σ (DN)	Difference from Read Noise (DN)
1	11.977	2.340
2	11.872	2.235
3	11.835	2.198
4	11.893	2.256
5	11.915	2.278
6	11.829	2.192
7	11.825	2.188
8	11.964	2.327
9	11.773	2.136
10	11.925	2.288
11	12.063	2.426
12	11.792	2.155
13	11.814	2.177
14	11.960	2.323
15	11.873	2.236
16	11.765	2.128
17	12.082	2.445
18	11.769	2.132
19	11.820	2.183
20	11.876	2.239
Mean	11.881 ± 0.021	2.244 ± 0.021

Table 1: Standard deviation of each individual bias frame measured in AstroImageJ, and its difference from the read noise measured with the Python script (9.637 DN).

The standard deviations of the individual bias frames all came out around 11.88 DN, roughly 2.24 DN higher than the 9.64 DN read noise from the Python script (Table 1). This is because the two measurements aren't actually measuring the same thing. The python scripts compare each pixel only to itself across all 20 frames, so only the random read noise is left. The standard deviation of a single frame compares every pixel to every other pixel, so it picks up both the read noise and the fixed pattern in the bias. Since independent noise adds in quadrature, the fixed pattern contributes about

$$\sigma_{\text{fixed}} = \sqrt{11.88^2 - 9.64^2} \approx 6.9 \text{ DN}. \tag{1}$$

The read noise itself is seemingly very uniform across the detector, since the full-frame and ROI measurements agree to within 0.001 DN. On the other hand, the bias level clearly varies across the chip. The master bias (Figure 2) shows a vertical gradient from brighter at the top to darker at the bottom, two dark columns, and a brighter strip along the left edge.



(a) Master bias.

(b) Bias frame after subtracting the master.

Figure 2: The fixed pattern in the bias (left) and the uniform noise left after removing it (right).

The gradient likely comes from how the chip is read out, since pixels farther from the output amplifier or read at a different time can sit at a slightly different bias level. The dark columns are likely just defects where those columns transfer charge differently than their neighbors. Because these features are fixed, subtracting a master bias removes them, which is why the bias-subtracted frames look so uniform.

1.3 Subtracting Bias From Bias Frames

As shown in Figure 2 we now have what we need to subtract the bias from our bias frames. Using AstroImageJ again, we can now get a new standard deviation for these new images.

Frame	σ (DN)	Frame	σ (DN)
1	9.870	11	9.983
2	9.773	12	9.771
3	9.770	13	9.758
4	9.778	14	9.956
5	9.852	15	9.765
6	9.790	16	9.760
7	9.791	17	10.046
8	9.924	18	9.760
9	9.758	19	9.772
10	9.961	20	9.872
Mean σ: 9.836 ± 0.020 DN		(scatter: 0.092 DN)	

Table 2: Standard deviation of each bias-subtracted bias frame, measured in AstroImageJ as directed in step 7.

Through Table 2 we can see the huge improvement in the standard deviation in terms of

lining up with the original standard deviation we got from the python code. Where we were once seeing a difference of ≈ 2.24 DN we now see one of ≈ 0.2 DN.

1.4 Using Two Bias Frames to Find Read Noise

For the last method, we take two bias frames, and using the Image Calculator in AstroImageJ we can subtract one bias frame from the other, and since the bias level and fixed pattern are the same in both frames, this gives us an image with near 0 bias. However, the read noise in each frame is random and adds in quadrature, so the noise in the difference image is

$$\sigma_{\text{diff}} = \sqrt{\sigma^2 + \sigma^2} = \sqrt{2} \cdot \sigma_{\text{RN}}. \tag{2}$$

The difference image gave us back a standard deviation of 14.25 DN, and using Equation 2 we see that the readout noise for this method is

$$\sigma_{\text{RN}} = \frac{14.25}{\sqrt{2}} \approx 10.08 \text{ DN}. \tag{3}$$

1.5 Summary of Readout Noise

Using the gain we measure in Section 2 ($g = 1.201 \pm 0.003$ e^-/DN), we converted each read noise measurement to electrons (Table 3).

Method	Read Noise (DN)	Read Noise (e^-)
Python script (full frame)	9.6370 ± 0.0005	11.57 ± 0.03
Python script (ROI)	9.636 ± 0.014	11.57 ± 0.03
Bias-subtracted frames	9.836 ± 0.020	11.81 ± 0.04
Difference of two bias frames	10.080 ± 0.002	12.11 ± 0.03

Table 3: Summary of read noise measurements, converted to electrons using $g = 1.201 \pm 0.003$ e^-/DN.

The uncertainties on the Python and two-frame values are purely statistical and are very small since they average over millions of pixels. The spread between the methods themselves (about 0.4 DN) is a better sense of how well we actually know the read noise.

1.6 Finding The Dark Current

We took a series of dark frames at a fixed CCD temperature of $\approx 7\,^\circ\text{C}$ with exposure times from 0.1 seconds to 60 seconds, plus a set of 5 minute darks at temperatures from $+7$ to $-35\,^\circ\text{C}$. For each frame we subtracted the master bias and took the median of the image as the dark level in DN. The dark charge per pixel is

$$D = g\,(N - N_0), \tag{4}$$

where N is the bias-subtracted median, N_0 is a constant offset (discussed below), and $g = 1.201 \pm 0.003$ e^-/DN is the gain from Exercise B.

One thing to note however is that the master bias was taken at $-34.6\,^\circ\text{C}$, while the time series darks were taken at $+7\,^\circ\text{C}$. Since the bias level shifts with temperature, subtracting

a cold master bias from warm darks leaves a leftover offset. That's why the fit to the time series has a negative intercept of $N_0 = -8.2$ DN instead of passing through zero. For the time series, we treat N_0 as a free parameter of the fit, so it doesn't affect the dark current.

1.6.1 Dark Current vs. Exposure Time

t (s)	T (°C)	N (DN)	$\tilde{N} - N_0$ (DN)
0.1	6.97	−8.10	0.14
10	6.97	−4.70	3.54
20	7.03	−1.00	7.24
30	6.97	2.60	10.84
40	6.97	6.20	14.44
50	7.00	10.05	18.29
60	6.94	13.50	21.74

Table 4: Dark level for the $\approx 7\,^{\circ}\mathrm{C}$ time series. N is the median after subtracting the master bias, and $N - N_0$ removes the −8.24 DN offset left by the master bias being taken at a colder temperature.

The negative values of N at short exposures aren't negative dark current. The camera's bias level is about 8 DN lower at $7\,^{\circ}\mathrm{C}$ than at $-34.6\,^{\circ}\mathrm{C}$, where the master bias was taken: the raw median of the 0.1 s frame is 1019 DN, compared to 1026.9 DN for the master bias. Subtracting the colder master bias therefore overshoots by ∼8 DN, and short exposures haven't built up enough dark charge to cover it. Fitting a line to N vs. exposure time (Figure 3) gives an intercept of $N_0 = -8.24$ DN, which is this offset, and a slope of 0.3629 ± 0.0019 DN/s. Since the offset only shifts the intercept, it doesn't affect the slope. Multiplying by the gain, the dark current at $7\,^{\circ}\mathrm{C}$ is

$$I = g \times \text{slope} = 0.436 \pm 0.003 \text{ e}^- \text{ pixel}^{-1} \text{ s}^{-1}. \tag{5}$$

Once the offset is removed, the dark charge grows linearly with exposure time and goes to zero at $t = 0$, as expected. As a check, the fit predicts 100.6 DN for the 300 s dark at $6.97\,^{\circ}\mathrm{C}$, and we measured 100.4 DN.

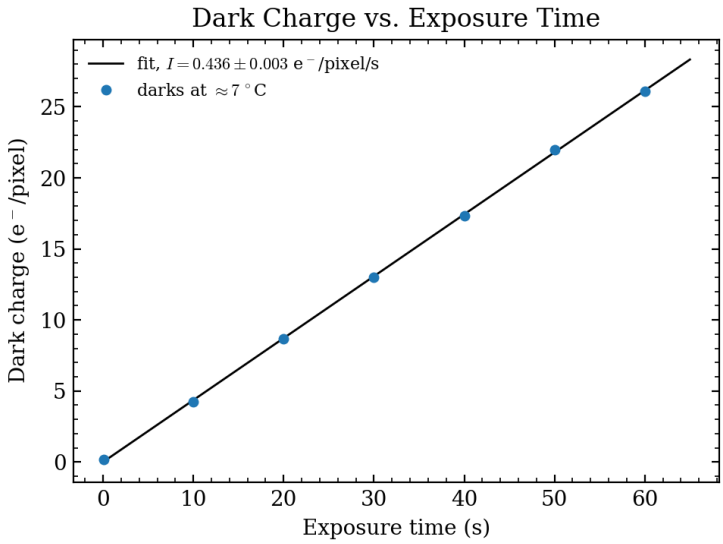


Figure 3: Dark charge vs. exposure time at $\approx 7\,^\circ$C, with the offset N_0 removed. The black line is the linear fit.

1.6.2 Dark Current vs. Temperature

For the 5 minute darks, the leftover bias offset depends on temperature. The four coldest frames (-16.9 to $-34.6\,^\circ$C) have an expected dark signal of $\lesssim 2$ DN, so their medians (-7.4 to -1.8 DN) are basically just the offset. We used them, plus the -8.24 DN intercept from the time series, to set the offset at each temperature, then subtracted it from the three warmer frames (Table 5). The dark current in the cold frames isn't zero, but it is too small to detect in a 5 minute exposure, so they aren't included in the fit.

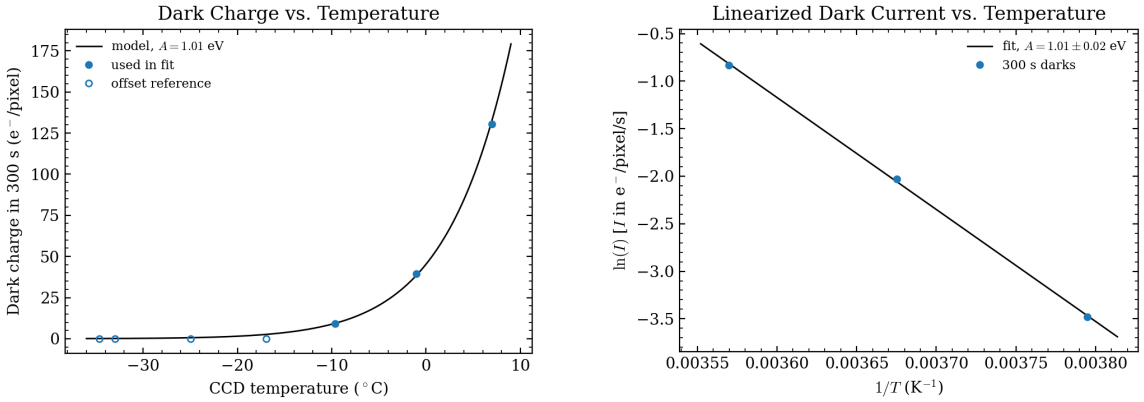
T (°C)	N (DN)	Signal (DN)	I (e$^-$/pixel/s)
6.97	100.40	108.64	0.435
-1.06	24.80	32.76	0.131
-9.62	0.05	7.71	0.031

Table 5: Dark current in the 5 minute darks with a detectable dark signal. N is the median after subtracting the master bias, and the signal has the temperature-dependent offset removed.

Now, looking at the equation $I = C\,e^{-A/kT}$, where A is the activation energy and $k = 8.617 \times 10^{-5}$ eV/K. Taking the log turns this into a line, $\ln I = \ln C - \frac{A}{k}\frac{1}{T}$, so fitting $\ln I$ vs. $1/T$ gives a slope of $-A/k$. Using the three warmest frames, we find a slope of $-11{,}780 \pm 210$ K, which gives

$$A = 1.01 \pm 0.02\,(\text{stat}) \pm 0.12\,(\text{sys})\ \text{eV}. \tag{6}$$

This is consistent with the band gap of silicon (1.12 eV), which is what you'd expect if the dark current comes from electrons being thermally kicked across the band gap. Figure 4a shows the dark charge vs. temperature with the model, and Figure 4b shows the linearized fit.



(a) (b)

Figure 4: (a) Dark charge in 5 minutes vs. CCD temperature, with the exponential model from the fit. Open circles are the cold frames used as offset references. (b) $\ln I$ vs. $1/T$ for the three frames with measurable dark current. The slope gives $A = 1.01 \pm 0.02$ eV (statistical).

1.6.3 Uncertainties

The statistical uncertainties come from the fit covariance, with the gain uncertainty added in quadrature for I. The dominant error is systematic and comes from the master bias being taken at a different temperature ($-34.6\,^\circ$C) than most of the darks. This leaves a temperature-dependent offset in every bias-subtracted dark, and we had to model it. For the time series this doesn't matter, since the offset just becomes the intercept of the fit. For the 300 s darks it does. The offset is roughly flat at ≈ -8 DN from -25 to $+7\,^\circ$C, but jumps to -1.75 DN in the coldest frame, so it isn't a simple function of temperature. If we instead draw a straight line for the offset from the $7\,^\circ$C intercept to the coldest frame, we get $A = 1.13$ eV. We take the 0.12 eV difference as the systematic uncertainty. The fit also uses only three points. A master bias taken at each temperature, or longer exposures at the cold temperatures, would remove most of this uncertainty.

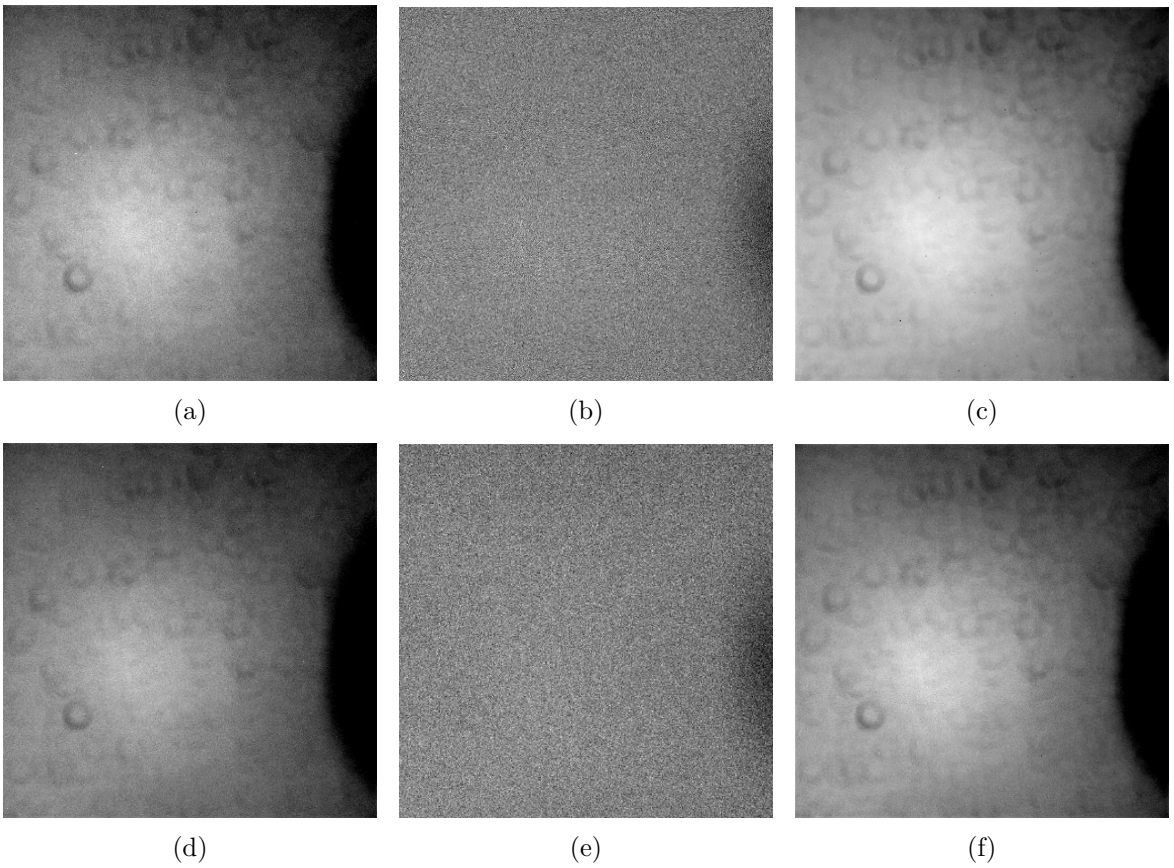
2 Exercise B

2.1 Introduction

Knowing the gain of a CCD is useful because it lets us convert the counts the camera records (DN) into the actual number of electrons, and therefore photons, that hit each pixel. The goal of this exercise was to measure the gain of the CCD, and to see how the noise in an image changes with the amount of light it receives. To do this, we took dome flats and built a photon transfer curve (PTC) from them. We also took twilight sky flats, which we use to compare the illumination of the two kinds of flats, and in Section 2.5 we explain why we adopt the gain from the dome flats.

For the dome flats, we set the filter to Johnson R, used the Astro 320 Star preset, and set the binning to 2. We raised the LED voltage from 16.85 V to 20.85 V and took continuous 6 second exposures while adjusting the voltage until the counts settled around 4000 DN. We then cooled the sensor to -30°C and took 24 dome flats at 0.2 seconds, followed by 24 dome

flats at 10 seconds. We also tested saturation by lowering the voltage 0.1 V at a time and taking one exposure at each step. For the sky flats, we started by pointing the telescope at the eastern sky (azimuth 90°, altitude 30°) with Simple Auto Dark on, but we also made use of the western sky to take sky flats near the end of twilight. We took 39 sky flats of 6 s each as the sky darkened, all well below saturation (Figure 7d shows a typical one, at ∼6,000 DN).



(a) (b) (c)

(d) (e) (f)

Figure 5: Example dome flats. The left column (a, d) shows 10 s exposures, the middle column (b, e) shows 0.2 s exposures, and the right column (c, f) shows the varying-exposure set at 80 s and 24 s.

2.2 Measuring the Gain

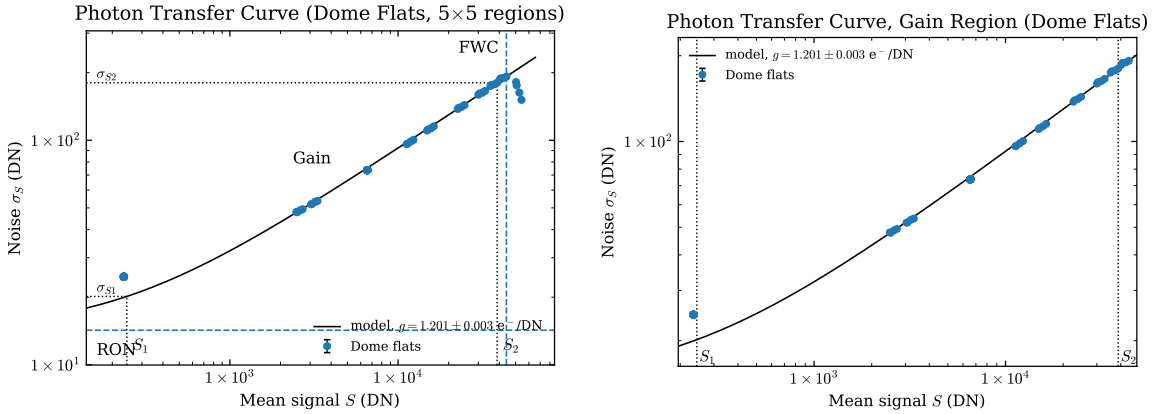
For a flat field, the noise in each pixel comes from photon shot noise plus read noise. In DN, the variance is

$$\sigma_S^2 = \frac{S}{g} + 2\sigma_{RN}^2, \tag{7}$$

where S is the mean signal and the factor of 2 comes from MaxIm dark subtracting our flats, which adds read noise a second time. To measure σ_S, we wrote a program that divides each flat into 5×5 pixel regions and calculates the mean and standard deviation inside each region. Regions with an unusually large standard deviation (from cosmic rays or hot pixels) were thrown out, and the rest were averaged to give one signal S and one noise σ_S for each image. For the varying-exposure flats, we split each image into four quadrants and did this for each quadrant separately, giving more points along the curve. Fitting Equation 7 to the points below 40,000 DN then gives the gain.

2.3 Photon Transfer Curves

Fitting Equation 7 to the dome flats gives $g = 1.201 \pm 0.003$ e$^-$/DN (Figure 6a). We adopt the dome flat gain g, as our final value for their advantages over sky flats covered in Section 2.5. Along with the 0.2 s and 10 s sets, the varying-exposure flats cover signals from about 230 DN up to 54,000 DN. Up to about 40,000 DN, the noise follows the expected $\sqrt{S}$ shot noise behavior (Figure 6b). Above that, the noise peaks at about 192 DN at a signal of 44,000 DN and then drops even though the signal keeps increasing, which is the detector reaching full well.



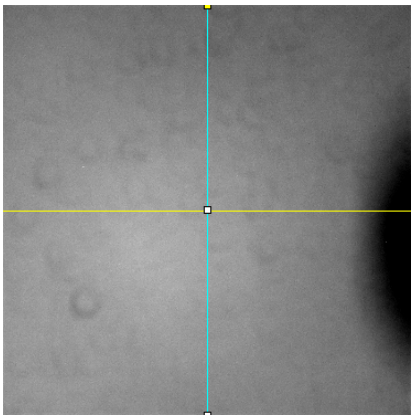
(a) Full range, showing the read noise floor, the gain region, and the drop at full well.

(b) Zoom on the gain region used for the fit centered at the region with useful points.

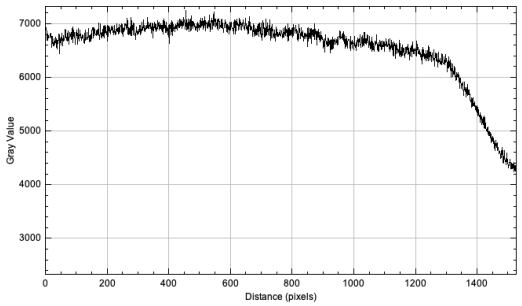
Figure 6: Photon transfer curve for the dome flats, measured from 5×5 pixel regions.

2.4 1-D Illumination Plots

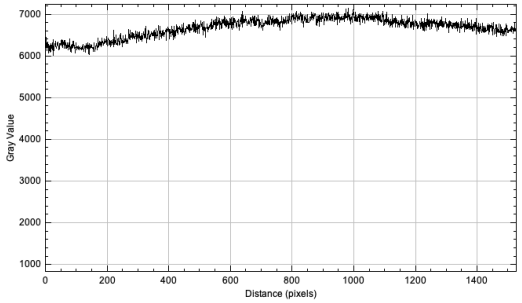
Figure 7 shows 1-D profiles across the middle of a dome flat and a sky flat. Both flats show the same drop off on the horizontal profile, which is expected due to that dark region on the right. This feature is likely vignetting, since we see this in every image taken. The faint ring-shaped donuts visible in both images are dust rings on the filter or camera window, and they also appear in the same places in both flats.



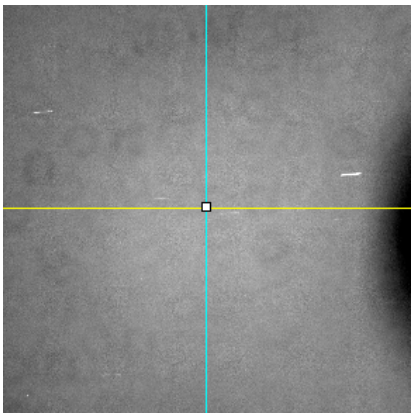
(a) Dome flat with the profile lines marked.



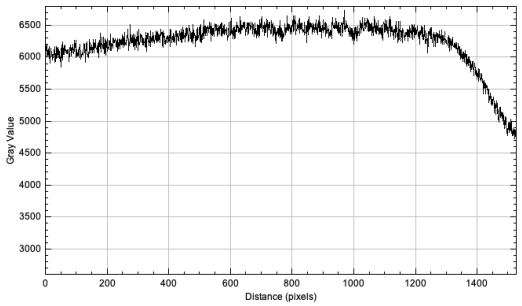
(b) Dome flat, horizontal profile.



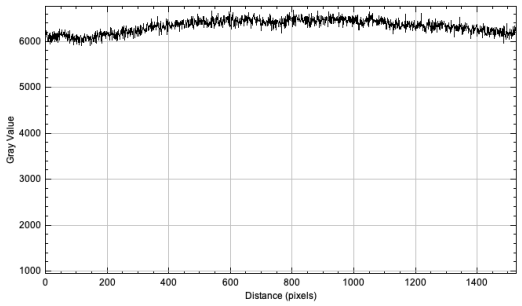
(c) Dome flat, vertical profile.



(d) Sky flat with the profile lines marked.



(e) Sky flat, horizontal profile.



(f) Sky flat, vertical profile.

Figure 7: 1-D illumination profiles of a 10 s dome flat (top) and a sky flat (bottom).

2.5 Comparing Sky and Dome Flats

Both kinds of flats have their own strengths. Sky flats are lit by the twilight sky, so light enters the telescope the same way it does for our science images, and their illumination ends up more uniform, as seen in Figure 7. However, the sky keeps darkening while we shoot, so every frame is at a slightly different level and its gradient shifts across the field. They're also affected by clouds, stars, and satellites (visible as the bright streaks in Figure 7d), and we only get a short window each night to take them. Dome flats, on the other hand, use a stable, controlled lamp, so we can take as many as we want at any exposure time, any time of night, with no stars or clouds to worry about. The downside is that the dome screen isn't lit perfectly evenly, which shows up as the larger brightness gradient in our dome flat. This being said, when talking about measuring the gain, stability matters most, which is why we adopt the dome flats value.

2.6 Saturation

The last question is which part of the camera saturates first. This is what our photon transfer curve answers (Figure 6a). Up to about 40,000 DN, the noise grows like $\sqrt{S}$. The noise then goes on to peak at about 192 DN at a signal of 44,000 DN, and in our brightest flat it drops to about 151 DN even though the mean signal keeps rising to about 54,000 DN. This drop happens because pixels start filling up. Once a pixel is full, it can't collect any more charge, so its value stops changing and the pixel to pixel scatter shrinks.

This turnover happens well below the ADC limit of 65,535 DN, so the ADC is not what saturates first. If the ADC were the limit, every pixel would hit the same maximum value and the signal would flatten out at one level. The same would be true if the analog electronics (the output amplifier) saturated first, the signal would be cut off at a single output level, below the ADC limit, and the noise would collapse suddenly rather than gradually. Instead, what we see from Figure 6 is that the mean signal keeps increasing while the noise drops, which fits pixels reaching their full well one at a time, since not every pixel holds exactly the same amount of charge. We therefore conclude that the CCD full well saturates first, at about 44,000 DN, or $\approx 53{,}000$ e^- using our gain.

3 Exercise C

3.1 Introduction

When measuring the brightness of a star, the quality of that measurement depends on more than just the star itself. If the telescope is out of focus, the star's light is spread over more pixels, each adding its own sky and read noise. If the exposure is too long, the brightest pixels fill up and the counts no longer follow the star's true brightness. Even with a sharp, unsaturated image, a single measurement does not tell us how much we can trust it. The point of this exercise was to test each of these: we found the best focus position, looked at how the FWHM and count rate of a star change with exposure time once it saturates, and checked how repeatable our photometry is by measuring the same star in a series of identical exposures. We began by stepping the focuser from 17,000 to 14,000 μm while imaging Omicron Aquilae, then took exposures of the TYC 4004-1801-1 field ranging from 5 to 180 s. On a second night, we took 10 back-to-back 10 s exposures of the same field, all with 2×2 binning.

3.2 Calibration

Following Photometry 101, every image was bias-subtracted and flat-fielded in AstroImageJ's CCD Data Processor. The master flat is the median of the 24 ten-second Johnson R dome flats from Exercise B, normalized to a mean of 1 (it drops to ≈ 0.6 at the vignetted right edge). We didn't take bias or dark frames at 2×2 binning, so we averaged the Exercise A master bias over 2×2 blocks using Python to match our images. That master bias was taken at $-34.6\,^{\circ}\mathrm{C}$, while these images were taken between 0 and $-15\,^{\circ}\mathrm{C}$, so it also leaves a small leftover offset of a few DN, and a bad column on the left side of the chip is only partly removed. Neither matters for our measurements: the offset is removed along with the sky by the background annulus during photometry, and the bad column doesn't cross any star we measured. We also didn't subtract a dark, since at these temperatures the dark current from Exercise A is only $\sim$0.1–0.6 e^{-}/pixel/s, and whatever is left is also removed by the background annulus.

3.3 FWHM vs. Focus Position

We began by drawing a line through o Aql in AstroImageJ and used *Plot Profile*. In one of the images taken, a charge leak runs through the top of the star, so we had to drop that image from the data.

Focus (μm)	FWHM (px)	FWHM ($''$)
17,000	12.5	15.5
16,500	10.2	12.6
16,000	8.5	10.5
15,500	8.5	10.5
15,000	7.9	9.7
14,500	14.0	17.3
14,000	14.7	18.2

Table 6: FWHM of o Aql at each focus position.

To get an uncertainty, we measured three images taken at the same focus (15,500 μm) and exposure time (1 s). Their standard deviation, ± 0.27 px, is the error bar on each point in Figure 8. It comes from where exactly the line crosses the star and from the pixel sampling. The pixel scale is $1.239''$/px from Exercise E.

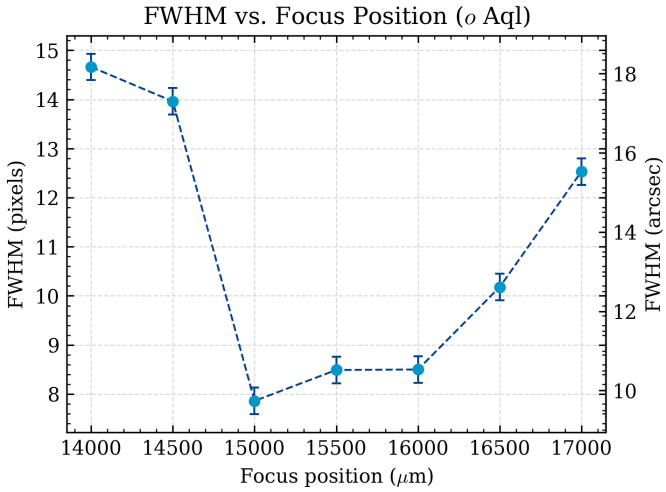


Figure 8: FWHM of *o* Aql vs. focus position. The error bars (±0.27 px) come from the scatter of repeated measurements at a fixed focus.

The FWHM forms a U/V shape: the star is smallest at 15,000 μm (7.9 px, $\approx 9.7''$) and gets wider the further the focus moves in either direction. Out of focus, the light from a star spreads over a disk instead of converging to a point. The size of the disk grows roughly in proportion to how far you are from focus, which gives the straight sides of the V shape. Near focus the FWHM levels off instead of dropping to zero, because it's limited by the seeing, the atmosphere blurs the star to a few arcseconds no matter how well the telescope is focused. Our minimum is larger than the $3.5''$ seeing from Exercise E for two reasons. Firstly, all of the *o* Aql images are saturated, which makes the star look wider than it really is (see Section 3.6), and secondly some images were taken at $-10\,^{\circ}$C while others were taken at $0\,^{\circ}$C, and the telescope's focus point shifts slightly as its structure changes temperature. That same effect is why the autofocus later settled on a different position (16,226 μm).

3.4 Repeatability of Photometry

To test how repeatable our photometry is, we took 10 back to back 10 s exposures of the TYC 4004-1801-1 field and measured the same star in each one with AstroImageJ's Multi-Aperture tool (aperture radius 15 px, sky annulus 25–60 px). Note that TYC 4004-1801-1 is different to the star mentioned in the manual (Ogma), due to every image having Ogma saturated even at extremely low exposure times. We picked a star near the center of the frame whose peak stayed below 41,000 DN in every frame, under the 44,000 DN where the CCD saturates (Section 2.6). The count rate in each image is

$$r = \frac{N}{t}, \tag{8}$$

where N is the sky-subtracted counts in the aperture and $t = 10$ s. Two frames were thrown out: Frame 7, where the star is smeared (width of 12.7 px compared to ∼5–6 px in the others, due to wind shaking the telescope, we can make this assumption by seeing every star is like this) and comes out 12% low, and Frame 10, which had a satellite crossing the field.

Frame	r (DN/s)	Deviation (%)
1	32,257	−0.44
2	32,303	−0.30
3	32,324	−0.24
4	32,091	−0.95
5	32,705	+0.94
6	32,486	+0.27
8	32,321	−0.24
9	32,713	+0.97

Table 7: Count rate of the repeatability star in each 10 s image and its deviation from the mean.

The mean count rate is $32{,}400 \pm 77$ DN/s (standard error of the mean), and the frame-to-frame scatter is 0.68%. Every frame lies within 1% of the mean.

In the case of whether uncertainties are dominated by photon statistics, we can say that they are not. Photon statistics predict an uncertainty of $\sigma_N = \sqrt{N}$, which AstroImageJ reports as 0.28% per frame. The real scatter between frames is 0.68%, so something other than photon noise is changing from frame to frame. The likely causes are changes in atmospheric transparency, and the star drifting slightly across pixels with small leftover flat-field errors.

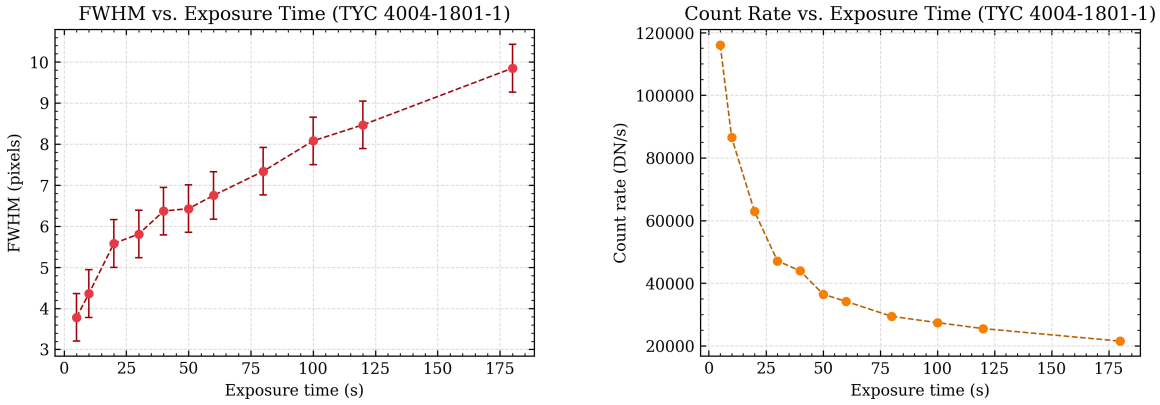
3.5 FWHM and Count Rate vs. Exposure Time

For the exposure series, we took images of the TYC 4004-1801-1 field with exposure times from 5 to 180 seconds and measured TYC 4004-1801-1 itself with Multi-Aperture (same radii as above). This star is already saturated in the 5 s image (peak of 45,600 DN), and stays saturated in every frame.

t (s)	Count rate (DN/s)	FWHM (px)	Peak (DN)
5	116,050	3.78	45,612
10	86,490	4.36	45,661
20	62,871	5.58	46,037
30	47,118	5.81	45,725
40	43,916	6.37	46,026
50	36,472	6.43	46,895
60	34,181	6.75	46,303
80	29,409	7.34	46,688
100	27,392	8.08	47,277
120	25,503	8.47	48,006
180	21,491	9.85	48,836

Table 8: Measurements of TYC 4004-1801-1 in the exposure series.

The FWHM grows steadily with exposure time, from 3.8 px at 5 s to 9.9 px at 180 s (Figure 9a), and the count rate drops from 116,000 to 21,500 DN/s (Figure 9b). Both happen for the same reason: the star is saturated. Its peak is stuck at ∼46,000–49,000 DN in every frame, so once the center of the star reaches full well, it can't collect any more charge. The extra light from a longer exposure only adds to the fainter wings of the star, which keeps growing outward, so the measured width grows. The total counts in the aperture still



(a) (b)

Figure 9: (a) FWHM and (b) count rate of TYC 4004-1801-1 vs. exposure time. The FWHM error bars (±0.58 px) are the scatter of AstroImageJ's FWHM in the repeatability images; the count-rate error bars (photon noise) are smaller than the points.

increase with exposure time, but much more slowly than t does, so dividing by a longer and longer t makes the count rate fall. By 180 s, the FWHM is also getting close to our 15 px aperture, so some light in the wings starts to fall outside it, which makes the drop a little steeper.

Now taking into consideration the question of 'does the exposure time affect the FWHM or the count rate?' For an unsaturated star, it really shouldn't affect either one. The FWHM is set by the seeing and focus, and a linear detector collects counts in proportion to the exposure time, so the count rate stays constant. We checked this with a fainter star in the same images that never saturates (peak below 32,000 DN even at 180 s): its count rate stayed at $1{,}505 \pm 23$ DN/s from 10 to 180 s and its FWHM stayed at ∼2.5–3 px. So exposure time seemingly only matters once a star saturates. After that, a longer exposure makes the star look wider and makes its measured count rate drop, which is why exposure times for photometry should be chosen to keep the stars of interest below full well.

3.6 Saturation

Figure 10 shows TYC 4004-1801-1 in the 180 s image. A profile drawn through the star shows that instead of a peaked shape, the top is flat at ∼48,200 DN across 7 pixels: those pixels have reached full well and can't record any more light.

4 Exercise D

4.1 Introduction

Measuring a star's brightness in a single filter tells us how bright it appears, but measuring it through several filters tells us much more. The difference between two magnitudes, called a color index (such as $V - R$), depends mostly on the star's surface temperature: hotter stars give off more of their light at shorter wavelengths, so they have smaller color indices than cooler stars. By measuring the colors of many stars in a cluster, we can tell different kinds of stars apart and see where they are in their lifecycles. The goal of the first part of



Figure 10: TYC 4004-1801-1 (circled in yellow) in the 180 s image.

this exercise was to do this for the open cluster NGC 6811. We took 8 exposures in each of Johnson V, R, and I, combined them into one image per filter, measured 10 stars in all three filters, and plotted the results on a color-color diagram and a color-magnitude diagram.

The second part of this exercise was astrophotography. By assigning a color to each filter and combining the three images, we can make a color image of an extended object and see how its color changes across it, the same temperature effect as in our photometry, applied to a whole galaxy. For this part, we imaged the Fireworks Galaxy (NGC 6946) in the same three filters and combined the images into a single color image.

4.2 Observations

To start, we looked at the open cluster NGC 6811 (RA $19^{\mathrm{h}}37.3^{\mathrm{m}}$, Dec $+46°23'$). With binning 2×2 and the CCD at $-10\,°\mathrm{C}$ and a focus position of 16,500 μm, we then took 8 exposures of 6 seconds in each of Johnson V, R, and I, aiming for peak counts around 30,000 DN so the stars stayed below full well.

4.3 Data Reduction

Each frame was calibrated as

$$\text{calibrated} = \frac{\text{raw} - \text{bias} - \text{dark}}{\text{flat}}, \tag{9}$$

using a Python script built on the results of Exercises A and B. We did not take bias or dark frames at 2×2 binning, so we used the master bias from Exercise A (binning 1), averaged over 2×2 blocks to match. For the dark, we used the dark-current law measured in Exercise A, $I(T) = I_0\, e^{-A/kT}$ with $A = 1.01$ eV, which gives only ≈ 0.03 e$^-$/pixel/s at $-10\,°\mathrm{C}$ (about 0.6 e$^-$ per binned pixel in a 6 s frame). Each filter was divided by its own normalized master flat, since vignetting and dust rings look different in each filter. The script also removed the hot pixels and bad columns that show up in the same place in every frame.

Following Photometry 101, we aligned the 8 calibrated frames in each filter in AstroImageJ (*Align stack using apertures*) and then summed them (*Process* → *Combine stack slices*, *Sum Slices*), giving one 48 s image per filter. Summing the frames boosts the signal-to-noise ratio by about $\sqrt{8} \approx 2.8$.

4.4 Photometry

We measured 10 stars in all three summed images with AstroImageJ's Multi-Aperture tool, clicking the stars in the same order in each filter (Figure 11). The stars have FWHM ≈ 3.4–3.7 px, and we used an aperture radius of 15 px with a sky annulus from 25 to 60 px, wide enough to catch all of each star's light. The instrumental magnitude of each star is

$$m_{\mathrm{inst}} = -2.5 \log_{10}\left(\frac{N}{t}\right), \tag{10}$$

where N is the sky-subtracted counts in the aperture and $t = 48$ s. The uncertainties come from AstroImageJ's photon-noise error on N, converted with $\sigma_m = 1.086\,\sigma_N/N$.

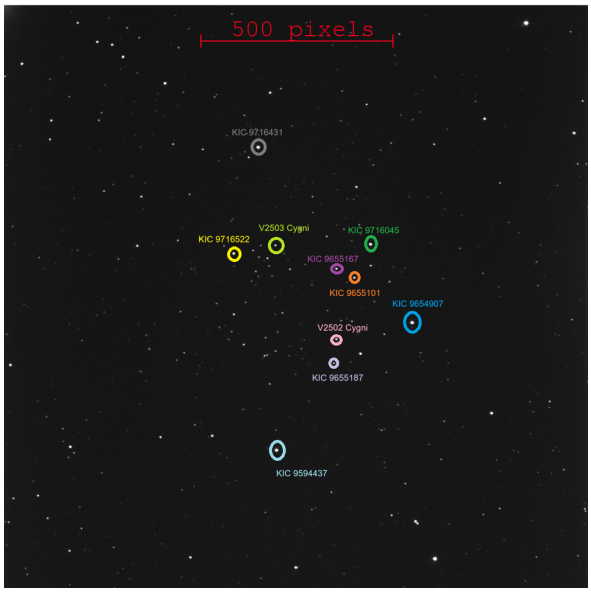


Figure 11: Above are all 10 stars used in order to perform photometry, each circle color represents the order they were measured along with their names. The order is as follows: Green, Yellow, Blue, Pink, Orange, Purple, Lime Green, Sky Blue, Grey, and Lavender.

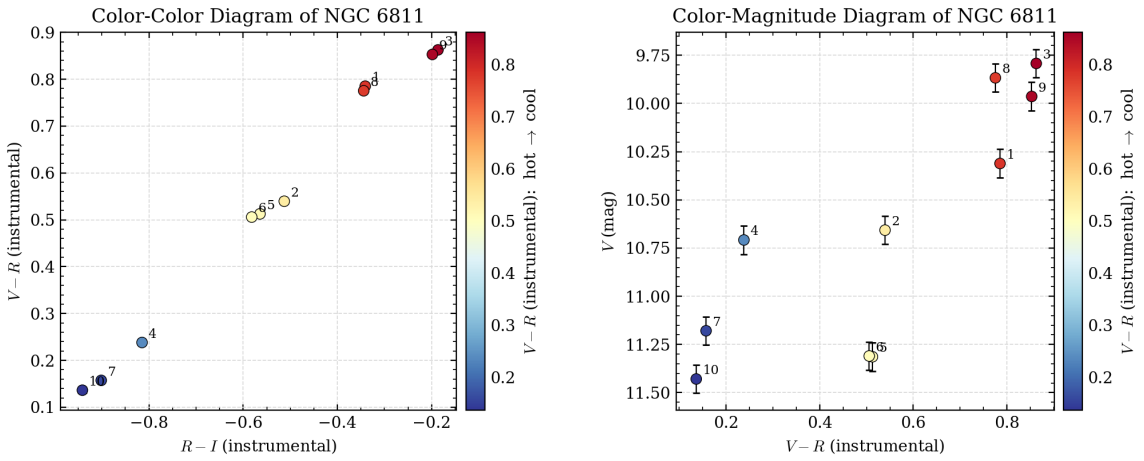
To put V on a real magnitude scale, we used catalog V magnitudes from SIMBAD [1] for the comparison stars. The median offset between catalog and instrumental V for the 7 non-variable comparison stars gives a zero point of 21.40 ± 0.07 mag. We left the two δ Scuti variables (V2502 Cyg and V2503 Cyg) out of the zero point since their brightness changes. As a check, the target star KIC 9716045 comes out at $V = 10.31$, compared to the catalog value of 10.30. For R and I, most of the comparison stars don't have catalog magnitudes, so we estimated their reference values from their Gaia colors using the transformations in Riello et al. (2021) [2]. The same median-offset method then gives zero points of 21.43 ± 0.07 mag in R and 20.47 ± 0.08 mag in I.

#	Star	V	R	I
1	KIC 9716045	10.31 ± 0.07	9.56 ± 0.07	8.94 ± 0.08
2	KIC 9716522	10.66 ± 0.07	10.15 ± 0.07	9.70 ± 0.08
3	KIC 9654907	9.79 ± 0.07	8.96 ± 0.07	8.19 ± 0.08
4	V2502 Cyg	10.71 ± 0.07	10.50 ± 0.07	10.35 ± 0.08
5	KIC 9655101	11.31 ± 0.07	10.83 ± 0.07	10.44 ± 0.08
6	KIC 9655167	11.31 ± 0.07	10.84 ± 0.07	10.46 ± 0.08
7	V2503 Cyg	11.18 ± 0.07	11.05 ± 0.07	10.99 ± 0.08
8	KIC 9594437	9.87 ± 0.07	9.12 ± 0.07	8.51 ± 0.08
9	KIC 9716431	9.96 ± 0.07	9.14 ± 0.07	8.38 ± 0.08
10	KIC 9655187	11.43 ± 0.07	11.33 ± 0.07	11.30 ± 0.08

Table 9: Calibrated V, R, and I magnitudes of the 10 measured stars in NGC 6811. Each band is calibrated with the median offset of the 7 non-variable comparison stars; the uncertainties are dominated by that zero point (photon noise contributes $\lesssim 0.01$ mag).

4.5 Results

Figure 12a shows the color-color diagram ($V - R$ vs. $R - I$) and Figure 12b shows the color-magnitude diagram (V vs. $V - R$). In the color-color diagram, all 10 stars fall along a single, nearly straight sequence. In the color-magnitude diagram they split into two groups: faint blue stars (#4, 7, 10) and bright red stars (#1, 3, 8, 9), with #2, 5, and 6 in between.



(a) (b)

Figure 12: (a) Color-color diagram of NGC 6811 in instrumental colors. (b) Color-magnitude diagram, with each star colored by its $V - R$ from blue (hot) to red (cool).

4.6 Discussion

A star's color mostly tells you its surface temperature, hotter stars put out more of their light at shorter (bluer) wavelengths, so they have smaller $V - R$ and $R - I$. Since both colors are driven by the same thing, stars land on one sequence in the color-color diagram, going from hot (bottom left) to cool (top right). That's what we see. The color-color diagram also doesn't depend on distance, since distance makes a star fainter in every filter by the same amount and the colors cancel it out.

One limitation is that $V - R$ vs. $R - I$ can't easily separate the effect of interstellar dust from temperature. Dust makes stars look redder and moves them along a direction almost parallel to the sequence. Diagrams using $U - B$ vs. $B - V$ are better for that, since the Balmer jump in hot stars bends the sequence away from the reddening direction. With only V, R, and I, our diagram is mostly a temperature sequence.

The color-magnitude diagram shows more structure, since now brightness is included. The faint blue stars (#4, 7, 10) are hot A/F stars near the cluster's main-sequence turnoff. Two of them, V2502 Cyg and V2503 Cyg, are δ Scuti pulsators, which sit in the instability strip right around this temperature. The brightest stars (#1, 3, 8, 9) are also the reddest, which only makes sense if they're giants: they're cool, but so large that they outshine the hotter main-sequence stars. For a $\sim$1 Gyr cluster like NGC 6811, stars just above the turnoff mass have already left the main sequence and become red giants, so this split is what we'd expect. With only 10 fairly bright stars, though, we only see the top of the turnoff and the giants, not the main sequence itself.

4.7 Uncertainties

The photon-noise errors on our colors are tiny ($\lesssim 0.01$ mag) because the stars are bright and we summed 8 frames, but they are a lower limit. In Exercise C, the real frame to frame scatter was about 2.4$\times$ the photon noise, from changes in seeing and atmospheric transparency, so a more realistic color uncertainty is $\sim$0.01–0.02 mag. The V magnitudes are limited by the zero point ($\pm$0.07 mag), which comes from the scatter in the catalog magnitudes of the comparison stars. Some of those catalog values are only given to 0.01–0.1 mag, and for star #5 the SIMBAD value ($V = 10.97$) disagrees with both its colors and our measurement ($V = 11.31$). We also used approximate bias and dark corrections, since no matching 2×2 calibration frames were taken, and the reddest star (#3) peaks at $\sim$41,000 DN per frame in I, close to full well, so its I magnitude may be slightly too faint.

4.8 Astrophotography

For the second part of this exercise, we imaged the Fireworks Galaxy (NGC 6946, RA $20^{\mathrm{h}}34.9^{\mathrm{m}}$, Dec $+60°09'$), taking 8 exposures of 15 s in each of Johnson V, R, and I with binning 2×2 and the CCD at $-10\,°$C. The handout calls for Nebulosity 4 to process these images, but since we no longer had access to it, we did everything in AstroImageJ instead.

Each filter was calibrated the same way as the NGC 6811 frames, with its own master flat. Within each filter, we aligned the 8 frames (*Process* $\rightarrow$ *Align stack using apertures*) and combined them by taking the median (*Process* $\rightarrow$ *Combine stack slices*, *Median*). We used the median here instead of summing the frames, because cosmic rays, satellite trails, and hot pixels show up in only one or two frames and the median throws them out, which is what we want for a clean picture. Next, we put the three combined images into a single stack (*Image* $\rightarrow$ *Stacks* $\rightarrow$ *Images to Stack*), aligned them with each other the same way, and split them back into three aligned images.

To make the color image, we adjusted the histogram at the bottom of AstroImageJ's image window, setting the black level at the sky and the white level only slightly above it so the faint spiral arms would show up. We then converted each image to 8-bit and combined them with *Image* $\rightarrow$ *Color* $\rightarrow$ *Merge Channels*, using I as red, R as green, and V as blue (Figure 13). Finally, we adjusted each color channel until the background looked neutral.



Figure 13: The Fireworks Galaxy (NGC 6946) made in AstroImageJ from 8 × 15 s exposures in each filter (I = red, R = green, V = blue).

The galaxy is a face-on spiral. Its core is the brightest part and looks yellow-orange, and the spiral arms can be traced around it, but they're much dimmer and have a brownish color rather than standing out in a different color from the core. In typical images of spiral galaxies, the arms look blue, because that's where new stars are forming and they're lit up by young, hot stars. We don't see that here, for a few reasons. Our bluest filter is V, so we're missing the blue and ultraviolet light where those hot stars are brightest, and the color differences we can pick up between V, R, and I are small. Since the arms are only slightly brighter than the sky, our short 15 s exposures leave them noisy, and stretching them that hard seems to have washed out whatever color difference is there. The field stars around the galaxy do show a clear range of colors, from blue-white to orange-red, which is the same temperature effect we measured in the NGC 6811 photometry.

The image also shows the limits of our data. The arms are only slightly brighter than the sky, so they only show up after a strong stretch, and even then the outer parts of the galaxy fade into the noise. NGC 6946 also sits close to the plane of the Milky Way, so its light is dimmed and reddened by dust in our own galaxy. There are also a few artifacts: the background has a slight blue-pink tint from imperfect color balancing, and the right edge of the frame has a bright glow where the flat field over-corrects the vignetted edge of the detector.

4.9 Changes to the Observing Procedure

For the photometry, we would take bias and dark frames at the same binning and temperature as the science frames, measure many more (and fainter) stars so the main sequence shows up in the color-magnitude diagram, and avoid variable stars such as V2502 Cyg and V2503 Cyg as comparison stars. We also did not notice until afterward that we had used Johnson I instead of Johnson B, so we would take a B image instead, which would let us make a $B - V$ diagram and compare to standard cluster CMDs. Finally, we would use a shorter exposure in I for the reddest giants. For the astrophotography, we'd point out that MaxIm cycled the filter after every image instead of finishing one filter first, so we'd set up the sequence properly before starting.

5 Exercise E

5.1 Introduction

Every measurement we make on an image is in pixels, but what we usually care about is how large something appears on the sky. The plate scale, the number of arcseconds of sky that each pixel covers, is what connects the two, and it lets us turn any distance on the CCD into an angle, such as the separation between two stars, the size of a star's image, or the size of the whole field of view. The goal of this exercise was to measure the plate scale of our camera using double stars. Since the angular separation of each pair is already listed in star catalogs, measuring how many pixels apart the two stars appear tells us how many arcseconds each pixel covers. We imaged ten double stars on 10/08/2026 and used the plate scale to find the typical seeing and the field of view of the camera.

5.2 Target Selection and Observations

Before observing, we selected candidate double stars in two ways. Our first attempts to build a list in AstroPlanner were unsuccessful because the software repeatedly crashed, so we chose pairs from the Washington Double Star Catalog (WDS) [3], accessed through the Stelle Doppie database [4], selecting pairs with separations from about $14''$ to $208''$ and small magnitude differences so that both stars could be well exposed without saturating. We later generated a second target list with AstroPlanner on the library computers.

At the telescope, the catalog-selected pairs produced clean, round images, while most of the AstroPlanner pairs appeared elongated, or "football-shaped." Every star in those frames was stretched in the same way, which points to telescope motion from the wind rather than a problem with the stars themselves. However, the effect appeared only on the AstroPlanner targets and not on our catalog pairs, which possibly suggests that it depended on where the telescope was pointing, for example lower altitude or facing the wind. We therefore used catalog pairs for nine of our ten measurements. The one AstroPlanner pair we kept, HR 8384, is also an outlier in our plate-scale measurement (Section 5.4).

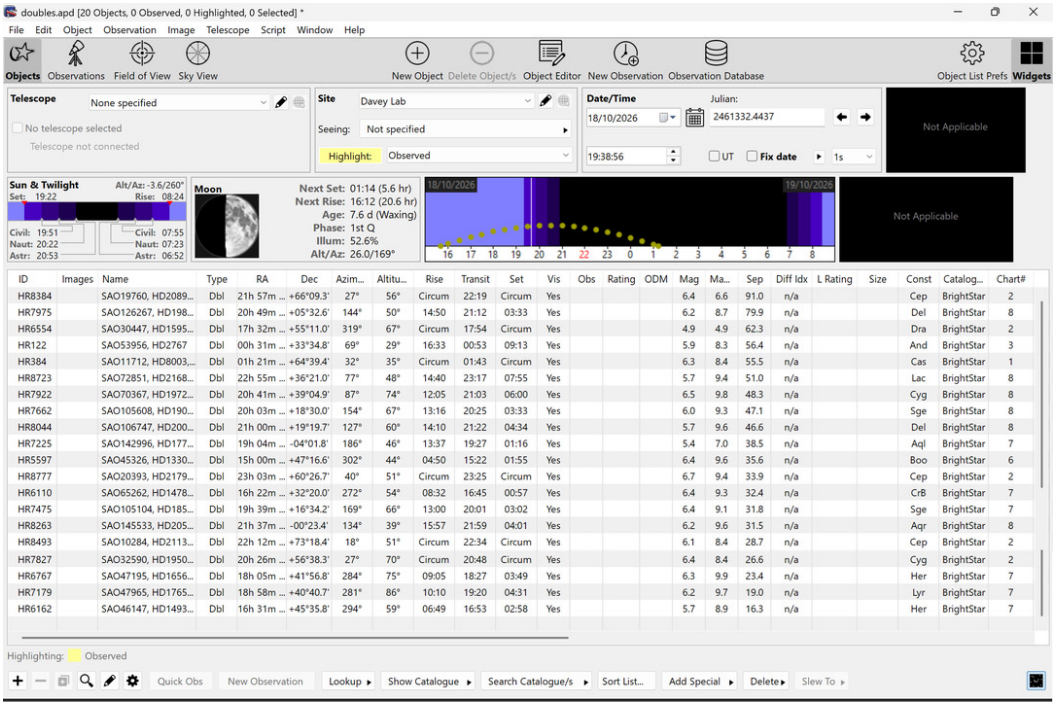
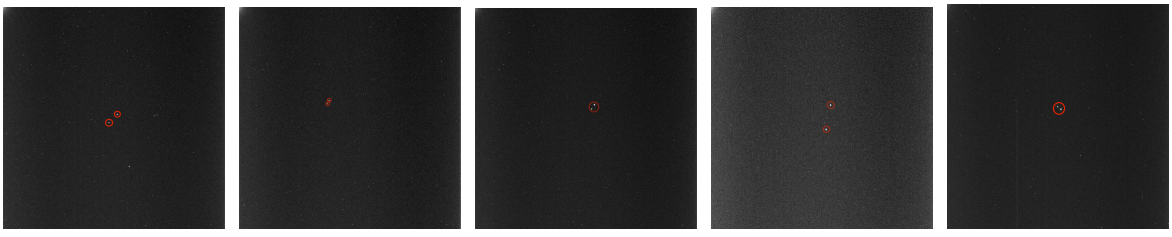
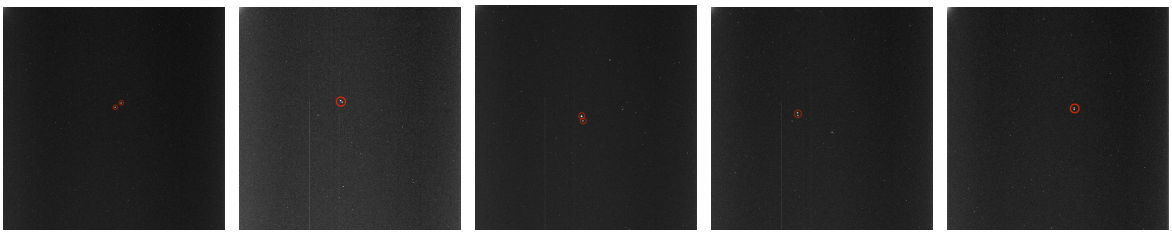


Figure 14: AstroPlanner screenshot of candidates selected.

All images were taken with Simple Auto Dark enabled, 2×2 binning, the *OIII* filter, and the Astro 320 Star preset. Example images of the pairs are shown in Figure 15.



(a) HR 8384 (b) 61 Cygni (c) ζ Lyrae (d) ϵ Lyrae (e) β Cygni



(f) ν Draconis (g) 40/41 Draconis (h) δ Cephei (i) 8 Lacertae (j) 100 Herculis

Figure 15: Images of the ten double stars observed for the plate-scale measurement. The red circles represent the binary observed

Table 10: Plate scale and star FWHM measured from each double star. HR 8384 (marked with *) was discarded as an outlier and not used in the mean plate scale.

Pair	Sky Sep. (″)	CCD Sep. (px)	Plate Scale (″/px)	FWHM_A (px)	FWHM_B (px)
HR 8384*	91	80.71	1.128	2.75	2.77
61 Cygni	32	25.79	1.241	2.68	2.80
ζ Lyrae	43	35.09	1.225	2.91	2.77
ϵ Lyrae	208	168.11	1.237	2.69	3.13
β Cygni	34.3	27.77	1.235	3.34	2.95
ν Draconis	61.6	49.88	1.235	2.73	2.68
40/41 Draconis	19	14.88	1.277	2.69	2.42
δ Cephei	41	32.66	1.256	3.46	2.86
8 Lacertae	22	17.96	1.225	2.81	2.75
100 Herculis	14	11.51	1.216	2.61	2.40
Mean			1.239 ± 0.006	2.81 ± 0.26	

5.3 Measuring the Star Separations

For each double star, we used Multi-Aperture in AstroImageJ to find the centroid (x, y) of both stars, with centroiding turned on so the aperture recentered on the peak of each star. The separation on the CCD is then

$$distance = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}, \tag{11}$$

and the plate scale for each pair is the catalog separation on the sky divided by *distance*. All images were taken with 2×2 binning, so the plate scale is per binned pixel. As a check that we measured the right stars, we compared the brightness difference between the two stars, $\Delta m = 2.5 \log_{10}(F_1/F_2)$, to the catalog magnitudes. Every pair matched to within about 0.15 mag, except δ Cephei, which is a variable star.

5.4 Plate Scale

Figure 16 shows the separation on the CCD against the separation on the sky. The pairs fall on a straight line through the origin, so the results are consistent. Nine of the ten pairs give plate scales between 1.216 and 1.277 ″/px. However, HR 8384 gave 1.128 ″/px, which is about 6σ below the mean of the others. HR 8384 is the only pair whose catalog separation came from AstroPlanner, while the rest came from other star catalogs, so the problem is most likely the catalog value rather than our measurement, and we discarded it. Averaging the remaining nine pairs gives a mean plate scale of

$$1.239 \pm 0.006 \text{ ″/px}, \tag{12}$$

where the uncertainty is the standard error of the nine values,

$$\sigma_{\bar{p}} = \frac{s}{\sqrt{N}} = \frac{0.018 \text{ ″/px}}{\sqrt{9}} \approx 0.006 \text{ ″/px}, \tag{13}$$

with s the standard deviation of the individual plate scales and $N = 9$ the number of pairs. Since the images are binned 2×2, each unbinned pixel covers

$$p_{\text{unbinned}} = \frac{p_{\text{binned}}}{2} = \frac{1.239 \text{ ″/px}}{2} \approx 0.62 \text{ ″/px}. \tag{14}$$

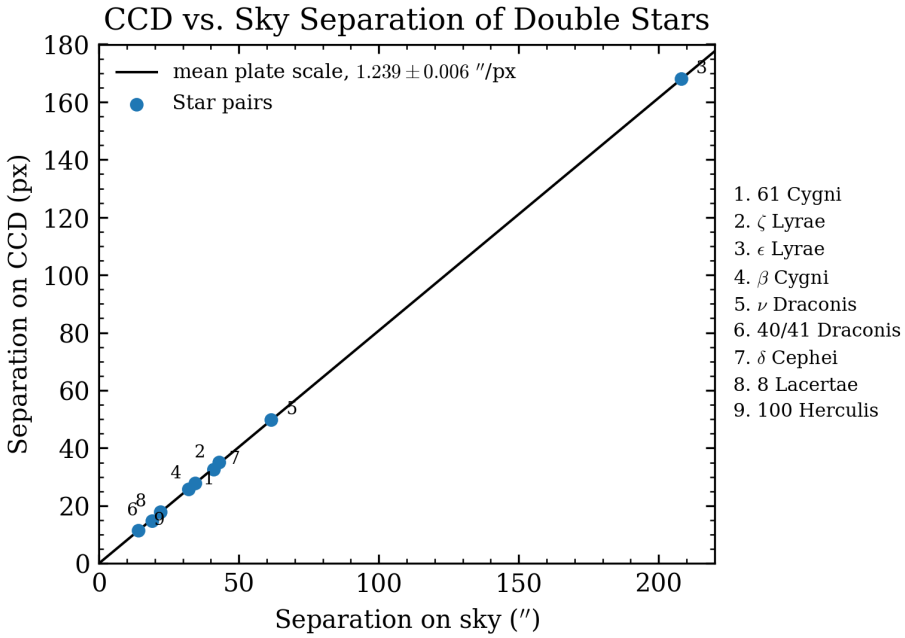


Figure 16: Separation on the CCD vs. separation on the sky for the nine double stars used in the mean. The line is our mean plate scale.

5.5 Uncertainties

The biggest source of uncertainty is the catalog separations themselves. Many are only given to two significant figures (e.g. 14″ and 19″), so for the closest pairs, rounding can change the plate scale by a significant amount. This is likely why 40/41 Draconis comes out the highest. Wide pairs like ϵ Lyrae (208″) are barely affected, which makes them our most reliable measurements. Some of these stars are also binaries that move over time, like 61 Cygni, so their true separation may differ slightly from the catalog value. The centroid of each star is good to about 0.1 px, which matters most for the closest pairs. For 100 Herculis, at only 11.5 px apart, that is about a 1% effect, and the close pairs also sit inside each other's background annulus. ϵ Lyrae is itself a "double double," with each star a tight $\sim$2″ pair, which made both stars slightly elongated in our images. Finally, the brighter star of δ Cephei peaked near 49,000 DN, close to where the CCD stops responding linearly. That affects its measured brightness but not its centroid.

5.6 Seeing and Field of View

The average FWHM of the 20 stars we measured was 2.81 ± 0.26 px. Multiplying by the plate scale gives a typical seeing of

$$2.81 \text{ px} \times 1.239 \text{ }''/\text{px} \approx 3.5 \pm 0.3 \text{ }''. \tag{15}$$

With 2×2 binning, the image is 1528 pixels across, so the field of view is

$$\frac{1528 \text{ px} \times 1.239 \text{ }''/\text{px}}{60} \approx 31.5 \pm 0.2 \text{ }', \tag{16}$$

or about half a degree, which matches the expected field of view for this camera.

References

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